

italics definition
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$$\text{Hom}(A \otimes B \otimes C, D) = \text{span} \left\{ \begin{array}{c} A \quad B \quad C \\ \diagdown \quad \diagup \quad \diagup \\ v_1 \bullet \quad \quad \bullet v_2 \\ \quad \quad \diagdown \quad \diagup \\ \quad \quad i \quad \quad \\ \quad \quad \quad \quad \quad D \end{array} \right\}$$

$$\begin{array}{c}
A \quad B \quad C \\
\diagdown \quad \diagup \quad \diagup \\
\bullet \quad \quad \bullet \\
\diagup \quad \diagdown \quad \diagdown \\
\mu \quad \nu \quad D
\end{array}
= \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(e\mu\nu)}^{(f\alpha\beta)}
\begin{array}{c}
A \quad B \quad C \\
\diagdown \quad \diagup \quad \diagup \\
\bullet \quad \bullet \\
\diagup \quad \diagdown \quad \diagdown \\
i \quad v_1 \quad v_2 \quad D
\end{array}$$

Diagram illustrating the construction of the NC state from the 0 and 1 states. The top row shows $|0\rangle$ (two small ellipses labeled 1 and τ inside a large ellipse labeled τ), $|1\rangle$ (two small ellipses labeled τ and τ inside a large ellipse labeled τ), and $|NC\rangle$ (two small ellipses labeled τ and τ inside a large ellipse labeled 1). The bottom row shows a single large ellipse labeled 1 containing three small ellipses labeled τ , with lines connecting them to the $|0\rangle$ and $|1\rangle$ states above.

$$\begin{array}{ccccc}
(A \otimes B) \otimes C & \xrightarrow{\alpha_{ABC}} & A \otimes (B \otimes C) \\
\downarrow \text{Hom}(-, D) & \text{Yoneda-lifting} & \downarrow \text{Hom}(-, D) \\
\text{Hom}((A \otimes B) \otimes C, D) & \cong & \text{Hom}(A \otimes (B \otimes C), D) \\
\downarrow \text{semisimplicity} & \text{semisimplicity} & \downarrow \text{semisimplicity} \\
\bigoplus_c \text{Hom}(A \otimes B, c) \otimes \text{Hom}(c \otimes C, D) & \cong & \bigoplus_{d \in \text{Irr}(\text{Cat})} \text{Hom}(A \otimes d, D) \otimes \text{Hom}(B \otimes C, d) \\
\downarrow & & \downarrow \\
\bigoplus_c V_c^{AB} \otimes V_D^{cC} & \text{choice of basis} & \bigoplus_c V_D^{Ac} \otimes V_c^{BC} \\
\downarrow & & \downarrow \\
|(A, B); c; \mu\rangle \otimes |c, C; D; \nu\rangle & \xlongequal{\hspace{1cm}} & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(c\mu\nu)}^{(f\alpha\beta)} |A, f; D; \beta\rangle \otimes |B, C; f; \alpha\rangle
\end{array}$$

$$\begin{array}{ccc}
 (A \otimes B) \otimes C & \xrightarrow{\alpha_{ABC}} & A \otimes (B \otimes C) \\
 \downarrow \text{Hom}(-, D) & & \downarrow \text{Hom}(-, D) \\
 \text{Hom}((A \otimes B) \otimes C, D) & \xrightarrow{\alpha_{A, B, C}} & \text{Hom}(A \otimes (B \otimes C), D)
 \end{array}$$

associator induces isomorphisms under yoneda

$$\begin{array}{ccc}
(A \otimes B) \otimes C & \xrightarrow[\sim]{\alpha_{ABC}} & A \otimes (B \otimes C) \\
\downarrow \text{Hom}(-, D) & & \downarrow \text{Hom}(-, D) \\
\text{Hom}((A \otimes B) \otimes C, D) & \xrightarrow[\sim]{\alpha_{A, B, C}} & \text{Hom}(A \otimes (B \otimes C), D) \\
\downarrow \text{Yoneda} & \text{associator induces isomorphisms under Yoneda} & \downarrow \text{Yoneda} \\
\bigoplus_{c \in \text{Irr}(\text{CAT})} \text{Hom}(A \otimes B, c) \otimes \text{Hom}(c \otimes C, D) & \cong & \bigoplus_{d \in \text{Irr}(\text{Cat})} \text{Hom}(A \otimes d, D) \otimes \text{Hom}(B \otimes C, d) \\
\parallel & & \parallel \\
\bigoplus_e V_e^{AB} \otimes V_D^{eC} & \cong & \bigoplus_f V_D^{Af} \otimes V_f^{BC} \\
\cap & & \cap \\
|(A, B); e; \mu\rangle \otimes |e, C; D; \nu\rangle & \xrightarrow[\text{6j-symbols}]{F_D^{ABC}} & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(e\mu\nu)}^{(f\alpha\beta)} |A, f; D; \beta\rangle \otimes |B, C; f; \alpha\rangle
\end{array}$$

